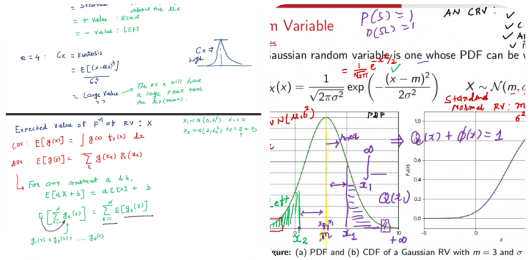
SCRIBE L8



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Generate a lecture scribe intended to serve as exam reference material. While writing the scribe, reason step by step through definitions, assumptions, proofs, derivations, and example solutions exactly as presented in the provided context. Include explicit reasoning in the scribe so that a student can follow the logical flow during revision. Use only the provided lecture slides and relevant textbook excerpts as context. Maintain a formal academic tone suitable for exam preparation. Follow the structure and scope of the lecture faithfully. Do not introduce any material, examples, explanations, or assumptions that are not present in the provided context. Prioritize correctness and faithfulness to the lecture material

CSE400 – Fundamentals of Probability in Computing

Lecture 8 Scribe: Gaussian Random Variable — Definitions, Properties, Standard Forms, and Applications

Instructor: Dr. Dhaval Patel

Course: CSE400 – Fundamentals of Probability in Computing

Purpose: Exam-reference lecture scribe prepared strictly from the provided lecture slides. All definitions, derivations, and example solutions follow the lecture flow so that students can reproduce each reasoning step during revision.

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1. Lecture Scope

This lecture focuses on:

1. Gaussian random variable — definition and properties.
2. Standard forms: Error function, Φ-function, and Q-function.
3. Relation between Φ and Q functions.
4. Evaluation of Gaussian CDF and tail probabilities.
5. Worked probability examples.
6. Application scenarios.
7. CDF-based problem solving involving mean, variance, skewness, and kurtosis.
8. Gaussian modeling and density estimation concepts.

The lecture develops Gaussian modeling as a practical framework for uncertainty and noise.

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2. Moments and Shape Measures of a Random Variable

The lecture briefly revisits higher-order standardized moments used to describe distribution shape.

2.1 Skewness

Skewness measures **symmetry** of a probability distribution about its mean.

For a random variable X,

$$c_3 = \frac{E[(X-\mu)^3]}{\sigma^3}$$

Interpretation:

- Positive value → distribution skewed to the **right**.
- Negative value → distribution skewed to the **left**.
- Zero → symmetric distribution.

Thus, skewness quantifies directional asymmetry around the mean.

2.2 Kurtosis

Kurtosis measures **peakedness** or tail heaviness:

$$c_4 = \frac{E[(X-\mu)^4]}{\sigma^4}$$

Interpretation:

- Larger value → sharper peak and heavier tails.

- Smaller value → flatter peak and lighter tails.

Thus, kurtosis indicates how concentrated probability mass is around the mean compared to the Gaussian reference.

---

3. Expected Value of a Function of a Random Variable

For a function $g(X)$:

Continuous Random Variable

$$E[g(X)] = \int g(x) f_X(x)\, dx$$

Discrete Random Variable

$$E[g(X)] = \sum g(x_k) P_X(x_k)$$

3.1 Linearity of Expectation

For constants a, b:

$$E[aX + b] = aE[X] + b$$

For sums:

$$E\left[\sum g_k(X)\right] = \sum E[g_k(X)]$$

This property allows decomposition of complicated expectations into manageable parts.

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4. Gaussian Random Variable

4.1 Definition

A random variable is **Gaussian** if its PDF is

$$f_X(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-m)^2}{2\sigma^2}\right)$$

Notation:

$$X \sim N(m, \sigma^2)$$

where

- m: mean,
- σ^2: variance.

4.2 Interpretation

Key observations:

- Distribution is symmetric about mean m.
- Most probability mass is concentrated near the mean.
- Spread is controlled by σ.

Graphs in lecture show:

- Bell-shaped PDF centered at m.
- Smooth S-shaped CDF.

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5. Standard Forms

To simplify calculations, Gaussian variables are standardized.

5.1 Error Function

$$\mathrm{erf}(x) = \frac{2}{\sqrt{\pi}} \int_0^x e^{-t^2} dt$$

Complementary form:

$$\mathrm{erfc}(x) = 1 - \mathrm{erf}(x)$$

5.2 Φ-Function (Standard Normal CDF)

For standard normal variable:

$$\Phi(x) = \int_{-\infty}^{x} \frac{1}{\sqrt{2\pi}} e^{-t^2/2} dt$$

This gives probability that standard normal ≤ x.

5.3 Q-Function (Tail Probability)

$$Q(x) = \int_x^\infty \frac{1}{\sqrt{2\pi}} e^{-t^2/2} dt$$

Represents **right-tail probability**.

5.4 Relationship

$$Q(x) = 1 - \Phi(x)$$

Thus,

- Φ → left-tail probability,
- Q → right-tail probability.

---

6. Evaluating Gaussian CDF

For $X \sim N(m, \sigma^2)$,

standardization:

$$Z = \frac{X - m}{\sigma}$$

Then,

$$F_X(x) = \Phi\left(\frac{x - m}{\sigma}\right)$$

Tail Probability

$$P(X > x) = Q\left(\frac{x - m}{\sigma}\right)$$

Lecture conclusion:

- Use **Φ** for CDF evaluation.
- Use **Q** for right-tail probabilities.

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7. Worked Probability Example

Given Gaussian PDF parameters, probabilities are converted into standard normal form.

Examples solved:

Example 1: $P(X < 0)$

Converted to standard form:

$$P(X < 0) = \Phi(\cdot)$$

Example 2: $P(X > 4)$

Expressed using Q-function:

$$P(X > 4) = Q(\cdot)$$

Example 3: $P(|X+3| < 2)$

Stepwise reasoning:

$$|X+3| < 2$$

$$-2 < X+3 < 2$$

$$-5 < X < -1$$

Probability becomes:

$$F_X(-1) - F_X(-5)$$

Using symmetry,

$$= 2\Phi(1) - 1 = 1 - 2Q(1)$$

Example 4: $P(|X-2| > 1)$

Equivalent to

$$X < 1 \quad \text{or} \quad X > 3$$

So,

$$P(X < 1) + P(X > 3)$$

expressed via Φ and Q functions.

---

8. Application Examples

Lecture applications include:

1. Thermal noise in electronic circuits.
2. Measurement error in sensors.
3. Packet delay variation in networks.

In each case,

Measured value = True value + Gaussian noise.

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9. CDF-Based Problem Solving

Given CDF:

$$F_X(x) = \begin{cases} 0, & x < 0 \\ x^2, & 0 < x < 1 \\ 1, & x > 1 \end{cases}$$

Step 1: PDF

Differentiate:

$$f_X(x) = 2x, \quad 0 < x < 1$$

Step 2: Mean

$$E[X] = \int_0^1 x(2x)dx$$

Step 3: Variance

Computed using:

$$Var(X) = E[X^2] - (E[X])^2$$

Step 4: Skewness and Kurtosis

Higher moments evaluated using definitions.

Lecture conclusion:

- Distribution is **left-skewed**.
- Distribution is **platykurtic** (lighter tails than Gaussian).

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10. Gaussian Modeling: Motivation

Lecture emphasizes:

Real measurements are noisy:

- Sensors fluctuate.
- Repeated measurements produce uncertainty cloud.

Model:

$$X = \mu + \sigma Z$$

with Z standard normal.

Probability describes uncertainty shape, not exact outcomes.

---

11. Density Estimation

In practice, parameters are unknown.

Steps:

1. Collect noisy samples.
2. Estimate mean and variance.
3. Fit Gaussian model.

Result:

Smooth probability model replaces raw histogram.

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12. Networking Application

Case study: Packet delay modeling.

Goals:

- Estimate typical delay.
- Measure jitter (variance).

Discussion point:

High variation harms real-time systems more than constant delay.

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13. Broader Computing Applications

Gaussian models appear in:

- Image denoising,
- Distributed system latency analysis,
- Robotics and sensor fusion,
- Autonomous vehicles.

Conclusion: **Uncertainty modeling is central to modern engineering systems.**

Final Exam Preparation Notes

Students should be able to:

1. Define Gaussian PDF and parameters.
2. Convert probabilities using standardization.

3. Use Φ and Q functions correctly.
4. Solve interval and tail probability problems.
5. Compute moments from PDFs/CDFs.
6. Interpret skewness and kurtosis.
7. Understand Gaussian modeling motivations.

If you want, next we can build a **one-page revision sheet** or **typical exam problems list** for fast mid-semester revision.

Generate a lecture scribe intended to serve as exam reference material. While writing the scribe, reason step by step through definitions, assumptions, proofs, derivations, and example solutions exactly as presented in the provided context. Include explicit reasoning in the scribe so that a student can follow the logical flow during revision. Use only the provided lecture slides and relevant textbook excerpts as context. Maintain a formal academic tone suitable for exam preparation. Follow the structure and scope of the lecture faithfully. Do not introduce any material, examples, explanations, or assumptions that are

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CSE400 – Fundamentals of Probability in Computing

Lecture 8 Scribe: Gaussian Random Variable — Definitions, Standard Forms, Evaluation, and Applications

Instructor: Dr. Dhaval Patel

Course: CSE400 – Fundamentals of Probability in Computing

Purpose: This lecture scribe is prepared strictly from the provided lecture slides and supporting material. It is intended to serve as exam-reference material. Definitions, derivations, and worked examples are written step by step so that students can reproduce reasoning during revision. The structure and scope follow the lecture faithfully, without introducing external material.

---

1. Lecture Scope

This lecture develops the Gaussian random variable and related evaluation tools. The progression in the lecture is:

1. Review of higher-order moments describing distribution shape.
2. Expected value of functions of random variables.
3. Definition and properties of the Gaussian random variable.
4. Standard forms: error function, Φ-function, and Q-function.
5. Relation between Φ and Q functions.
6. Evaluation of CDF and tail probabilities.

7. Worked probability examples.
8. Application examples.
9. Problem solving using a given CDF.
10. Gaussian modeling and density estimation concepts.

---

2. Shape Measures of a Distribution

The lecture revisits standardized moments that describe the shape of distributions.

2.1 Skewness

Skewness measures symmetry of a distribution about the mean.

It is defined as

$$c_3 = \frac{E[(X-\mu)^3]}{\sigma^3}$$

Interpretation:

- Positive skewness → distribution stretches more to the right.
- Negative skewness → distribution stretches more to the left.
- Zero skewness → symmetric distribution.

Thus, skewness measures directional asymmetry around the mean.

2.2 Kurtosis

Kurtosis measures how peaked or flat a distribution is compared to a Gaussian distribution.

It is defined as

$$c_4 = \frac{E[(X-\mu)^4]}{\sigma^4}$$

Interpretation:

- Larger value → sharper peak and heavier tails.
- Smaller value → flatter distribution with lighter tails.

Hence, kurtosis measures concentration around the mean.

---

3. Expected Value of a Function of a Random Variable

For a function $g(X)$, expectation is computed differently for discrete and continuous variables.

Continuous Case

$$E[g(X)] = \int g(x) f_X(x)\, dx$$

Discrete Case

$$E[g(X)] = \sum g(x_k) P_X(x_k)$$

Linearity of Expectation

For constants a and b:

$$E[aX + b] = aE[X] + b$$

For sums:

$$E\left[\sum g_k(X)\right] = \sum E[g_k(X)]$$

This property simplifies evaluation of expectations.

---

4. Gaussian Random Variable

4.1 Definition

A random variable is Gaussian if its PDF has the form

$$f_X(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-m)^2}{2\sigma^2}\right)$$

Notation:

$$X \sim N(m, \sigma^2)$$

where:

- m is the mean,
- σ^2 is the variance.

4.2 Observed Properties

From the lecture graphs:

- Distribution is symmetric about mean m.
- Maximum density occurs at $x = m$.
- Spread is controlled by σ.
- The CDF forms an S-shaped curve.

---

5. Standard Forms

To compute probabilities easily, Gaussian variables are standardized.

5.1 Error Function

Defined as

$$\mathrm{erf}(x) = \frac{2}{\sqrt{\pi}} \int_0^x e^{-t^2} dt$$

Complementary function:

$$\mathrm{erfc}(x) = 1 - \mathrm{erf}(x)$$

5.2 Φ-Function (Standard Normal CDF)

For a standard normal variable:

$$\Phi(x) = \int_{-\infty}^{x} \frac{1}{\sqrt{2\pi}} e^{-t^2/2} dt$$

This gives left-tail probability.

5.3 Q-Function (Gaussian Tail Function)

Defined as

$$Q(x) = \int_{x}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-t^2/2} dt$$

This gives right-tail probability.

5.4 Relationship Between Φ and Q

The lecture states:

$$Q(x) = 1 - \Phi(x)$$

Thus:

- Φ is convenient for CDF evaluation.
- Q is convenient for right-tail probabilities.

---

6. Evaluating Gaussian CDF

For $X \sim N(m, \sigma^2)$, define standard variable:

$$Z = \frac{X - m}{\sigma}$$

Then,

$$F_X(x) = \Phi\left(\frac{x - m}{\sigma}\right)$$

Tail Probability

Right-tail probability becomes:

$$P(X > x) = Q\left(\frac{x - m}{\sigma}\right)$$

The lecture emphasizes:

- Use Φ for CDF values.
- Use Q for right-tail computations.

---

7. Worked Probability Transformations

Several probability expressions are rewritten into standard form.

Example: Interval Probability

For

$$|X+3| < 2$$

Stepwise reasoning:

$$-2 < X+3 < 2$$

$$-5 < X < -1$$

Thus,

$$P(-5 < X < -1) = F_X(-1) - F_X(-5)$$

Using symmetry relations:

$$= 2\Phi(1) - 1 = 1 - 2Q(1)$$

Example: Two-Tail Probability

For

$$|X-2| > 1$$

Equivalent condition:

$$X < 1 \quad \text{or} \quad X > 3$$

Hence,

$$P(X < 1) + P(X > 3)$$

expressed using Φ and Q functions.

---

8. Application Examples

Gaussian modeling appears in:

- Thermal noise in circuits.
- Measurement error in sensors.
- Packet delay variation in networks.

In each case:

Measured value = True value + Gaussian noise.

---

9. Problem Solving Using Given CDF

Given CDF:

$$F_X(x) = \begin{cases} 0, & x < 0 \\ x^2, & 0 < x < 1 \\ 1, & x > 1 \end{cases}$$

Step 1: PDF

Differentiate:

$$f_X(x) = 2x, \quad 0 < x < 1$$

Step 2: Mean and Variance

Mean and variance are computed using definitions of expectation and moment relations.

Step 3: Skewness and Kurtosis

Using moment definitions, the lecture concludes:

- Distribution is left-skewed.
- Distribution is platykurtic (lighter tails than Gaussian).

---

10. Gaussian Modeling Motivation

Lecture motivation:

Real-world measurements are noisy.

Repeated observations form uncertainty clouds.

Model:

$$X = \mu + \sigma Z$$

where Z is standard normal.

Probability models uncertainty, not exact outcomes.

---

11. Density Estimation

In practice, parameters are unknown.

Procedure:

1. Observe noisy samples.
2. Estimate mean and variance.
3. Fit Gaussian PDF.

This produces a smooth predictive model from raw data.

---

12. Networking Application

Packet delay modeling example:

Objectives:

- Estimate typical delay.
- Measure delay variation (jitter).

Discussion: large variation damages real-time communication more than constant delay.

---

13. Broader Applications

Gaussian models appear in:

- Image denoising,
- Distributed system latency,
- Sensor fusion,
- Autonomous systems.

Conclusion: uncertainty modeling is fundamental across computing systems.

Final Exam Preparation Checklist

Students should be able to:

1. State Gaussian PDF and parameters.
2. Standardize variables.
3. Use Φ and Q functions correctly.
4. Evaluate tail and interval probabilities.
5. Compute moments from PDF/CDF.
6. Interpret skewness and kurtosis.
7. Explain Gaussian modeling motivation.

If desired, the next step can be a **one-page exam revision summary** or a **set of typical mid-sem problems with solutions** for fast preparation.

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CSE400 – Fundamentals of Probability in Computing

Lecture 8 Scribe: Gaussian Random Variable — Definition, Standard Forms, Probability Evaluation, and Applications

Instructor: Dr. Dhaval Patel

Course: CSE400 – Fundamentals of Probability in Computing

Lecture Topic: Gaussian Random Variable – Definition and Properties

Purpose: This lecture scribe is prepared strictly from the provided lecture slides and material and is intended to serve as exam-reference notes. All reasoning, derivations, and example solutions follow the lecture flow so that students can reproduce steps during revision. No additional assumptions or external material are introduced.

1. Lecture Scope

The lecture develops the Gaussian random variable and tools required to compute probabilities involving it. The sequence presented in the lecture is:

1. Review of higher-order moments describing distribution shape.
2. Expected value of functions of random variables.
3. Definition and properties of Gaussian random variables.
4. Standard forms: error function, Φ-function, and Q-function.
5. Relation between Φ and Q functions.
6. Evaluation of Gaussian CDF and tail probabilities.
7. Example probability evaluations.
8. Application examples.
9. Problem solving using a given CDF.
10. Motivation for Gaussian modeling and density estimation.
11. Applications in network systems and broader computing contexts.

The lecture emphasizes probabilistic modeling of uncertainty.

---

2. Shape Measures of a Random Variable

The lecture begins with measures describing distribution shape.

2.1 Skewness

Skewness measures symmetry of a distribution about its mean.

It is defined as

$$c_3 = \frac{E[(X-\mu)^3]}{\sigma^3}$$

Reasoning:

- The numerator measures asymmetry around the mean.
- Normalization by σ^3 makes the quantity dimensionless.

Interpretation:

- Positive value → right-skewed distribution.
- Negative value → left-skewed distribution.
- Zero → symmetric distribution.

2.2 Kurtosis

Kurtosis measures how peaked or flat a distribution is compared with a Gaussian distribution.

Definition:

$$c_4 = \frac{E[(X-\mu)^4]}{\sigma^4}$$

Reasoning:

- Fourth power emphasizes extreme deviations.
- Normalization makes comparison possible across distributions.

Interpretation:

- Larger values indicate heavier tails and sharper peaks.
- Smaller values indicate flatter distributions.

---

3. Expected Value of a Function of a Random Variable

For a function $g(X)$, expectation depends on variable type.

Continuous Random Variable

$$E[g(X)] = \int g(x) f_X(x)\, dx$$

Reasoning:

Each value contributes weighted by its probability density.

Discrete Random Variable

$$E[g(X)] = \sum g(x_k) P_X(x_k)$$

Each outcome contributes weighted by probability mass.

Linearity of Expectation

For constants a, b:

$$E[aX + b] = aE[X] + b$$

For sums:

$$E\left[\sum g_k(X)\right] = \sum E[g_k(X)]$$

Reasoning:

Expectation distributes over addition and constant scaling, simplifying calculations.

---

4. Gaussian Random Variable

4.1 Definition

A random variable is Gaussian if its PDF is

$$f_X(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-m)^2}{2\sigma^2}\right)$$

Notation:

$$X \sim N(m, \sigma^2)$$

where:

- m is the mean,
- σ^2 is the variance.

4.2 Observed Properties from Graphs

From lecture graphs:

- PDF is symmetric around mean m.
- Maximum occurs at mean.
- Spread increases with larger σ.
- CDF forms an S-shaped curve.

Thus, mean determines center, variance controls spread.

---

5. Standard Forms

To simplify probability evaluation, Gaussian variables are converted to standard form.

5.1 Error Function

Defined as

$$\mathrm{erf}(x) = \frac{2}{\sqrt{\pi}} \int_0^x e^{-t^2} dt$$

Complementary function:

$$\mathrm{erfc}(x) = 1 - \mathrm{erf}(x)$$

These arise in Gaussian integral evaluation.

5.2 Φ-Function (Standard Normal CDF)

For standard normal variable,

$$\Phi(x) = \int_{-\infty}^{x} \frac{1}{\sqrt{2\pi}} e^{-t^2/2} dt$$

This represents left-tail probability.

5.3 Q-Function (Tail Probability)

Defined as

$$Q(x) = \int_{x}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-t^2/2} dt$$

This represents right-tail probability.

5.4 Relationship Between Φ and Q

Lecture identity:

$$Q(x) = 1 - \Phi(x)$$

Thus,

- Φ is used for CDF evaluation,
- Q is natural for right-tail probabilities.

---

6. Evaluating Gaussian CDF

For $X \sim N(m, \sigma^2)$, define

$$Z = \frac{X - m}{\sigma}$$

Then

$$F_X(x) = \Phi\left(\frac{x - m}{\sigma}\right)$$

Reasoning:

Standardization converts any Gaussian into standard normal.

Tail Probability

$$P(X > x) = Q\left(\frac{x - m}{\sigma}\right)$$

Hence Q-function directly computes right-tail probabilities.

---

7. Example Probability Transformations

Example: Interval Condition

Given

$$|X+3| < 2$$

Stepwise reasoning:

$$-2 < X+3 < 2$$

Subtracting 3:

$$-5 < X < -1$$

Hence,

$$P(-5 < X < -1) = F_X(-1) - F_X(-5)$$

Using symmetry relation:

$$= 2\Phi(1) - 1 = 1 - 2Q(1)$$

Example: Two-Sided Condition

For

$$|X-2| > 1$$

Equivalent to

$$X < 1 \quad \text{or} \quad X > 3$$

Thus probability becomes

$$P(X < 1) + P(X > 3)$$

which is evaluated using Φ and Q functions.

---

8. Application Examples

Lecture examples include:

1. Thermal noise in electronic circuits.
2. Measurement error in sensors.
3. Packet delay variation in networks.

In each case:

Measured value = True value + Gaussian noise.

---

9. Problem Solving Using Given CDF

Given:

$$F_X(x) = \begin{cases} 0, & x < 0 \\ x^2, & 0 < x < 1 \\ 1, & x > 1 \end{cases}$$

Step 1: Obtain PDF

Differentiate:

$$f_X(x) = 2x, \quad 0 < x < 1$$

Step 2: Compute Mean and Variance

Mean and variance are obtained using expectation formulas.

Step 3: Skewness and Kurtosis

Moments are computed using definitions.

Lecture conclusion:

- Distribution is left-skewed.
- Distribution is platykurtic.

---

10. Gaussian Modeling Motivation

Lecture motivation:

Real systems exhibit randomness:

- Sensors fluctuate,
- Measurements vary,
- Data appears noisy.

Model:

$$X = \mu + \sigma Z$$

where Z is standard normal.

Thus probability describes uncertainty structure.

---

11. Density Estimation

In practice parameters are unknown.

Process:

1. Observe noisy data.
2. Estimate mean and variance.
3. Fit Gaussian model.

Result: smooth probabilistic representation replaces raw data histogram.

---

12. Networking Application

Packet delay modeling:

Objectives:

- Estimate mean delay,
- Measure jitter (variation).

Observation: large variation harms real-time applications more than constant delay.

13. Broader Applications

Gaussian models are used in:

- Image denoising,
- Distributed systems,
- Sensor fusion,
- Autonomous systems.

Conclusion: uncertainty modeling is fundamental in computing systems.

Final Revision Checklist

Students should be able to:

1. State Gaussian PDF.
2. Perform standardization.
3. Use Φ and Q correctly.
4. Evaluate tail and interval probabilities.
5. Compute moments from PDFs and CDFs.
6. Interpret skewness and kurtosis.
7. Explain Gaussian modeling motivation.

End of Lecture 8 Scribe.

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CSE400 – Fundamentals of Probability in Computing

Lecture 8 Scribe: Gaussian Random Variable — Definition, Standard Forms, Probability Evaluation, and Applications

Instructor: Dr. Dhaval Patel

Course: CSE400 – Fundamentals of Probability in Computing

Lecture Topic: Gaussian Random Variable — Definition and Properties

Purpose: This lecture scribe is prepared strictly from the provided lecture slides and supporting textbook excerpts. It is intended to serve as exam-reference material. Definitions, derivations, and example solutions are written step by step so that students can reproduce reasoning during revision. The structure, notation, and logical progression follow the lecture faithfully, and no additional assumptions or material are introduced.

1. Lecture Scope and Structure

The lecture focuses on Gaussian random variables and probability evaluation tools. The structure followed in the lecture is:

1. Review of standardized moments describing distribution shape.
2. Expected value of functions of random variables.
3. Definition and properties of Gaussian random variables.
4. Standard forms: error function, Φ-function, and Q-function.
5. Relation between Φ and Q functions.
6. Evaluation of Gaussian CDF and tail probabilities.
7. Worked probability examples.
8. Application examples.
9. Problem solving using a given CDF.
10. Gaussian modeling motivation and density estimation.
11. Applications in network systems and computing contexts.

The lecture builds the connection between probability theory and modeling uncertainty in practical systems.

---

2. Shape Measures of a Distribution

The lecture begins by recalling measures that describe distribution shape beyond mean and variance.

2.1 Skewness

Skewness measures symmetry of a probability distribution around the mean.

It is defined as

$$c_3 = \frac{E[(X-\mu)^3]}{\sigma^3}$$

Reasoning:

- The term $(X-\mu)^3$ preserves sign, so positive and negative deviations contribute differently.
- Dividing by σ^3 makes the measure dimensionless.

Interpretation:

- Positive skewness → distribution extends more to the right.
- Negative skewness → distribution extends more to the left.
- Zero skewness → symmetric distribution.

Thus skewness quantifies directional asymmetry.

2.2 Kurtosis

Kurtosis measures how peaked or flat the distribution is relative to the Gaussian case.

Definition:

$$c_4 = \frac{E[(X-\mu)^4]}{\sigma^4}$$

Reasoning:

- Fourth power emphasizes large deviations.
- Normalization removes dependence on scale.

Interpretation:

- Larger values imply heavier tails and sharper peak.
- Smaller values imply flatter distributions.

Hence kurtosis indicates concentration of probability around the mean.

---

3. Expected Value of a Function of a Random Variable

Expectation of a function depends on the type of random variable.

3.1 Continuous Random Variable

For a function $g(X)$,

$$E[g(X)] = \int g(x) f_X(x)\, dx$$

Reasoning:

Each value contributes proportionally to its density.

3.2 Discrete Random Variable

$$E[g(X)] = \sum g(x_k) P_X(x_k)$$

Each possible value contributes weighted by probability mass.

3.3 Linearity of Expectation

For constants a, b,

$$E[aX + b] = aE[X] + b$$

Also,

$$E\left[\sum g_k(X)\right] = \sum E[g_k(X)]$$

This property simplifies expectation calculations.

---

4. Gaussian Random Variable

4.1 Definition

A random variable is Gaussian if its probability density function is

$$f_X(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-m)^2}{2\sigma^2}\right)$$

Notation:

$$X \sim N(m, \sigma^2)$$

where:

- m is the mean,
- σ^2 is the variance.

4.2 Observed Properties

From lecture graphs:

- PDF is symmetric about mean m.
- Maximum density occurs at the mean.
- Larger variance produces wider spread.
- CDF has S-shaped growth from 0 to 1.

Thus mean sets location, variance sets spread.

---

5. Standard Forms

To evaluate probabilities conveniently, Gaussian variables are standardized.

5.1 Error Function

Defined as

$$\mathrm{erf}(x) = \frac{2}{\sqrt{\pi}} \int_0^x e^{-t^2} dt$$

Complementary error function:

$$\mathrm{erfc}(x) = 1 - \mathrm{erf}(x)$$

These functions appear in Gaussian integral evaluation.

5.2 Φ-Function (Standard Normal CDF)

Defined for standard normal variable as

$$\Phi(x) = \int_{-\infty}^x \frac{1}{\sqrt{2\pi}} e^{-t^2/2} dt$$

This gives probability to the left of x.

5.3 Q-Function (Gaussian Tail Function)

Defined as

$$Q(x) = \int_x^\infty \frac{1}{\sqrt{2\pi}} e^{-t^2/2} dt$$

This gives probability in the right tail.

5.4 Relation Between Φ and Q

Lecture identity:

$$Q(x) = 1 - \Phi(x)$$

Thus:

- Φ is used for CDF evaluation.
- Q is convenient for right-tail probabilities.

---

6. Evaluating Gaussian CDF

For $X \sim N(m, \sigma^2)$, define standardized variable

$$Z = \frac{X - m}{\sigma}$$

Then,

$$F_X(x) = \Phi\left(\frac{x - m}{\sigma}\right)$$

Reasoning:

Standardization converts the variable to standard normal form.

Tail Probability

Right-tail probability becomes

$$P(X > x) = Q\left(\frac{x - m}{\sigma}\right)$$

Hence Q-function directly evaluates right tails.

---

7. Worked Probability Transformations

Example: Interval Probability

Given

$$|X + 3| < 2$$

Step-by-step reasoning:

$$-2 < X + 3 < 2$$

Subtracting 3:

$$-5 < X < -1$$

Thus probability becomes

$$F_X(-1) - F_X(-5)$$

Using symmetry:

$$= 2\Phi(1) - 1 = 1 - 2Q(1)$$

Example: Two-Sided Condition

Given

$$|X - 2| > 1$$

Equivalent to

$$X < 1 \quad \text{or} \quad X > 3$$

Thus,

$$P(X < 1) + P(X > 3)$$

evaluated using Φ and Q.

---

8. Application Examples

Lecture examples:

1. Thermal noise in circuits.
2. Measurement error in sensors.
3. Packet delay variation in communication networks.

In all cases:

Measured value = True value + Gaussian noise.

9. Problem Solving Using Given CDF

Given CDF:

$$F_X(x) = \begin{cases} 0, & x < 0 \\ x^2, & 0 < x < 1 \\ 1, & x > 1 \end{cases}$$

Step 1: Obtain PDF

Differentiate:

$$f_X(x) = 2x, \quad 0 < x < 1$$

Step 2: Mean and Variance

Mean and variance are computed using expectation formulas.

Step 3: Skewness and Kurtosis

Higher moments are evaluated.

Lecture conclusion:

- Distribution is left-skewed.

- Distribution is platykurtic.

---

10. Gaussian Modeling Motivation

Lecture motivation:

Real-world measurements exhibit randomness.

Repeated observations produce uncertainty clouds.

Model:

$$X = \mu + \sigma Z$$

where Z is standard normal.

Probability models uncertainty rather than exact values.

---

11. Density Estimation

In practical systems parameters are unknown.

Procedure:

1. Collect noisy samples.
2. Estimate mean and variance.
3. Fit Gaussian model.

This replaces raw histogram data with smooth probabilistic representation.

---

12. Networking Application

Packet delay modeling:

Objectives:

- Estimate typical delay.
- Measure delay variation (jitter).

Observation: large variation harms real-time applications more than constant delay.

---

13. Broader Computing Applications

Gaussian modeling appears in:

- Image denoising,
- Distributed systems,
- Sensor fusion,
- Autonomous systems.

Conclusion: uncertainty modeling is central in modern computing systems.

Final Revision Checklist

Students should be able to:

1. State Gaussian PDF and parameters.
2. Standardize variables.
3. Use Φ and Q correctly.
4. Evaluate interval and tail probabilities.
5. Compute moments from PDF/CDF.
6. Interpret skewness and kurtosis.
7. Explain Gaussian modeling motivation.

End of Lecture 8 Scribe.

Generate a lecture scribe intended to serve as exam reference material. While writing the scribe, reason step by step through definitions, assumptions, proofs, derivations, and example solutions exactly as presented in the provided context. Include explicit reasoning in the scribe so that a student can follow the logical flow during revision. Use only the provided lecture slides and relevant textbook excerpts as context. Maintain a formal academic tone suitable for exam preparation. Follow the structure and scope of the lecture faithfully. Do not introduce any material, examples, explanations, or assumptions that are not present in the provided context. Prioritize correctness and faithfulness to the lecture material

CSE400 – Fundamentals of Probability in Computing

Lecture 8 Scribe: Gaussian Random Variable — Definition, Standard Forms, Probability Evaluation, and Applications

Instructor: Dr. Dhaval Patel

Course: CSE400 – Fundamentals of Probability in Computing

Lecture Topic: Gaussian Random Variable — Definition and Properties

Purpose:

This lecture scribe is prepared strictly from the provided lecture slides and supporting textbook excerpts and is intended to serve as exam-reference material. All definitions, assumptions, derivations, and example solutions are written step by step exactly following the lecture progression so that students can reproduce the reasoning during examination revision. No material beyond the lecture context is introduced.

---

1. Lecture Scope and Logical Flow

The lecture develops the Gaussian random variable and tools used to evaluate probabilities involving it. The structure of the lecture proceeds as follows:

1. Review of standardized higher-order moments describing distribution shape.
2. Expected value of functions of random variables.
3. Definition and properties of Gaussian random variables.
4. Standard forms: error function, Φ-function, and Q-function.
5. Relation between Φ and Q functions.

6. Evaluation of Gaussian CDF and tail probabilities.
7. Worked probability examples.
8. Application examples.
9. Problem solving using a given CDF.
10. Gaussian modeling motivation.
11. Density estimation from observed data.
12. Applications in networking and computing systems.

The lecture emphasizes how probability models uncertainty in engineering systems.

---

2. Measures Describing Distribution Shape

The lecture begins by recalling measures that describe distribution characteristics beyond mean and variance.

2.1 Skewness

Skewness measures symmetry of a probability distribution around the mean.

It is defined as

$$c_3 = \frac{E[(X-\mu)^3]}{\sigma^3}$$

Reasoning:

- The cubic term preserves sign, distinguishing deviations on either side of the mean.

- Normalization by σ^3 removes units.

Interpretation:

- Positive skewness → distribution extends further to the right.
- Negative skewness → distribution extends further to the left.
- Zero skewness → symmetric distribution.

Thus, skewness quantifies directional asymmetry.

2.2 Kurtosis

Kurtosis measures how peaked or flat the distribution is compared with a Gaussian distribution.

It is defined as

$$c_4 = \frac{E[(X-\mu)^4]}{\sigma^4}$$

Reasoning:

- Fourth power emphasizes large deviations.
- Normalization makes comparison independent of scale.

Interpretation:

- Larger values correspond to heavier tails and sharper peaks.
- Smaller values correspond to flatter distributions.

Thus kurtosis measures concentration around the mean.

3. Expected Value of a Function of a Random Variable

Expectation of a function depends on whether the variable is continuous or discrete.

3.1 Continuous Random Variable

For function $g(X)$,

$$E[g(X)] = \int g(x) f_X(x)\, dx$$

Reasoning:

Each value contributes weighted by probability density.

3.2 Discrete Random Variable

$$E[g(X)] = \sum g(x_k) P_X(x_k)$$

Each possible outcome contributes weighted by probability.

3.3 Linearity of Expectation

For constants a, b:

$$E[aX + b] = aE[X] + b$$

Also,

$$E\left[\sum g_k(X)\right] = \sum E[g_k(X)]$$

This property allows expectations of sums to be computed term by term.

---

4. Gaussian Random Variable

4.1 Definition

A random variable is Gaussian if its PDF is

$$f_X(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-m)^2}{2\sigma^2}\right)$$

Notation:

$$X \sim N(m, \sigma^2)$$

where

- m is the mean,
- σ^2 is the variance.

4.2 Observed Properties from Lecture Graphs

From lecture figures:

- Distribution is symmetric around mean m.
- Maximum density occurs at the mean.
- Spread increases as σ increases.
- CDF has smooth S-shaped growth.

Thus, mean determines location, variance controls spread.

---

5. Standard Forms Used in Gaussian Calculations

To simplify probability evaluation, Gaussian variables are standardized.

5.1 Error Function

Defined as

$$\mathrm{erf}(x) = \frac{2}{\sqrt{\pi}} \int_0^x e^{-t^2} dt$$

Complementary error function:

$$\mathrm{erfc}(x) = 1 - \mathrm{erf}(x)$$

These arise from Gaussian integrals.

5.2 Φ-Function (Standard Normal CDF)

Defined as

$$\Phi(x) = \int_{-\infty}^{x} \frac{1}{\sqrt{2\pi}} e^{-t^2/2} dt$$

This represents left-tail probability.

5.3 Q-Function (Gaussian Tail Function)

Defined as

$$Q(x) = \int_{x}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-t^2/2} dt$$

This gives right-tail probability.

5.4 Relation Between Φ and Q

Lecture identity:

$$Q(x) = 1 - \Phi(x)$$

Hence:

- Φ is convenient for CDF evaluation.
- Q is convenient for right-tail probabilities.

---

6. Evaluating Gaussian CDF

For $X \sim N(m, \sigma^2)$, define standardized variable

$$Z = \frac{X - m}{\sigma}$$

Then,

$$F_X(x) = \Phi\left(\frac{x - m}{\sigma}\right)$$

Reasoning:

Standardization converts any Gaussian variable into standard normal form.

Tail Probability

Right-tail probability becomes

$$P(X > x) = Q\left(\frac{x - m}{\sigma}\right)$$

Thus Q-function directly evaluates right-tail events.

---

7. Worked Probability Transformations

Example: Interval Probability

Given

$$|X+3| < 2$$

Step-by-step reasoning:

$$-2 < X+3 < 2$$

Subtracting 3:

$$-5 < X < -1$$

Hence,

$$P(-5 < X < -1) = F_X(-1) - F_X(-5)$$

Using symmetry relations:

$$= 2\Phi(1) - 1 = 1 - 2Q(1)$$

Example: Two-Sided Probability

Given

$$|X-2| > 1$$

Equivalent condition:

$$X < 1 \quad \text{or} \quad X > 3$$

Thus,

$$P(X < 1) + P(X > 3)$$

evaluated using Φ and Q functions.

---

8. Application Examples

Lecture applications include:

- Thermal noise in circuits,
- Measurement errors in sensors,
- Packet delay variations in networks.

In each case:

Measured value = True value + Gaussian noise.

---

9. Problem Solving Using a Given CDF

Given

$$F_X(x) = \begin{cases} 0, & x < 0 \\ x^2, & 0 < x < 1 \\ 1, & x > 1 \end{cases}$$

Step 1: Obtain PDF

Differentiate:

$$f_X(x) = 2x, \quad 0 < x < 1$$

Step 2: Mean and Variance

Mean and variance are computed using expectation formulas.

Step 3: Skewness and Kurtosis

Using moment definitions, lecture concludes:

- Distribution is left-skewed.
- Distribution is platykurtic.

---

10. Gaussian Modeling Motivation

Lecture motivation:

Real systems produce noisy measurements.

Repeated observations form uncertainty clouds.

Model:

$$X = \mu + \sigma Z$$

where Z is standard normal.

Thus probability models uncertainty, not exact outcomes.

---

11. Density Estimation

In practical situations parameters are unknown.

Procedure:

1. Observe noisy samples.
2. Estimate mean and variance.
3. Fit Gaussian distribution.

This produces a smooth probabilistic model from raw data.

---

12. Networking Application

Packet delay modeling aims to:

- Estimate mean delay,
- Measure jitter (variation).

Observation: large variation harms real-time applications more than constant delay.

---

13. Broader Computing Applications

Gaussian modeling is used in:

- Image denoising,
- Distributed systems,
- Sensor fusion,
- Autonomous systems.

Conclusion: uncertainty modeling is central across computing systems.

Final Revision Checklist

Students should be able to:

1. State Gaussian PDF and parameters.
2. Perform standardization.
3. Use Φ and Q functions correctly.
4. Evaluate interval and tail probabilities.
5. Compute moments from PDFs and CDFs.
6. Interpret skewness and kurtosis.
7. Explain Gaussian modeling motivation.

End of Lecture 8 Scribe.

Generate a lecture scribe intended to serve as exam reference material. While writing the scribe, reason step by step through definitions, assumptions, proofs, derivations, and example solutions exactly as presented in the provided context. Include explicit reasoning in the scribe so that a student can follow the logical flow during revision. Use only the provided lecture slides and relevant textbook excerpts as context. Maintain a formal academic tone suitable for exam preparation. Follow the structure and scope of the lecture faithfully. Do not introduce any material, examples, explanations, or assumptions that are not present in the provided context. Prioritize correctness and faithfulness to the lecture material

CSE400 – Fundamentals of Probability in Computing

Lecture 8 Scribe: Gaussian Random Variable — Definition, Standard Forms, Probability Evaluation, and Applications

Instructor: Dr. Dhaval Patel

Course: CSE400 – Fundamentals of Probability in Computing

Lecture Topic: Gaussian Random Variable — Definition and Properties

Purpose:

This lecture scribe is prepared strictly from the provided lecture slides and supporting textbook excerpts. It is intended to serve as exam-reference material. All definitions, derivations, and example solutions are written step by step exactly following the lecture

progression so that students can reproduce the reasoning during revision. No material beyond the lecture context is introduced.

---

1. Lecture Scope and Logical Flow

The lecture develops Gaussian random variables and probability evaluation methods. The progression followed in the lecture is:

1. Review of standardized higher-order moments describing distribution shape.
2. Expected value of functions of random variables.
3. Definition and properties of Gaussian random variables.
4. Standard forms: error function, Φ-function, and Q-function.
5. Relation between Φ and Q functions.
6. Evaluation of Gaussian CDF and tail probabilities.
7. Worked probability examples.
8. Application examples.
9. Problem solving using a given CDF.
10. Gaussian modeling motivation.
11. Density estimation from observed data.
12. Applications in networking and broader computing contexts.

The lecture emphasizes that probability models uncertainty present in engineering systems.

---

2. Measures Describing Distribution Shape

The lecture begins by revisiting measures describing distribution shape beyond mean and variance.

2.1 Skewness

Skewness measures asymmetry of a distribution about its mean.

It is defined as

$$c_3 = \frac{E[(X-\mu)^3]}{\sigma^3}$$

Reasoning

- The cubic term preserves sign, distinguishing deviations to the left and right.
- Normalization by σ^3 removes dependence on units.

Interpretation

- Positive skewness → distribution extends more to the right.
- Negative skewness → distribution extends more to the left.
- Zero skewness → symmetric distribution.

Thus skewness quantifies directional asymmetry.

2.2 Kurtosis

Kurtosis measures peakedness or flatness of the distribution relative to a Gaussian distribution.

Definition:

$$c_4 = \frac{E[(X-\mu)^4]}{\sigma^4}$$

Reasoning

- Fourth power emphasizes large deviations.
- Normalization removes scale dependence.

Interpretation

- Larger value → sharper peak and heavier tails.
- Smaller value → flatter distribution.

Thus kurtosis indicates concentration of probability around the mean.

---

3. Expected Value of a Function of a Random Variable

Expectation depends on whether the random variable is continuous or discrete.

3.1 Continuous Random Variable

For function $g(X)$,

$$E[g(X)] = \int g(x) f_X(x)\, dx$$

Each value contributes weighted by probability density.

3.2 Discrete Random Variable

$$E[g(X)] = \sum g(x_k) P_X(x_k)$$

Each possible outcome contributes weighted by probability mass.

3.3 Linearity of Expectation

For constants a, b:

$$E[aX + b] = aE[X] + b$$

Also,

$$E\left[\sum g_k(X)\right] = \sum E[g_k(X)]$$

This allows expectations of sums to be computed term by term.

---

4. Gaussian Random Variable

4.1 Definition

A random variable is Gaussian if its PDF is

$$f_X(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-m)^2}{2\sigma^2}\right)$$

Notation:

$$X \sim N(m, \sigma^2)$$

where:

- m is the mean,
- σ^2 is the variance.

4.2 Observed Properties

From lecture graphs:

- Distribution is symmetric around mean.
- Maximum density occurs at mean.
- Larger variance produces wider spread.
- CDF is S-shaped.

Thus mean sets location; variance sets spread.

---

5. Standard Forms for Gaussian Calculations

To evaluate probabilities conveniently, Gaussian variables are standardized.

5.1 Error Function

Defined as

$$\mathrm{erf}(x) = \frac{2}{\sqrt{\pi}} \int_0^x e^{-t^2} dt$$

Complementary function:

$$\operatorname{erfc}(x) = 1 - \operatorname{erf}(x)$$

5.2 Φ-Function (Standard Normal CDF)

Defined as

$$\Phi(x) = \int_{-\infty}^{x} \frac{1}{\sqrt{2\pi}} e^{-t^2/2} dt$$

This gives left-tail probability.

5.3 Q-Function (Gaussian Tail Function)

Defined as

$$Q(x) = \int_{x}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-t^2/2} dt$$

This gives right-tail probability.

5.4 Relation Between Φ and Q

Lecture identity:

$$Q(x) = 1 - \Phi(x)$$

Thus:

- Φ is used for CDF evaluation.
- Q is convenient for right-tail probabilities.

---

6. Evaluating Gaussian CDF

For $X \sim N(m, \sigma^2)$, define standardized variable

$$Z = \frac{X - m}{\sigma}$$

Then,

$$F_X(x) = \Phi\left(\frac{x - m}{\sigma}\right)$$

Standardization converts the variable into standard normal form.

Tail Probability

Right-tail probability:

$$P(X > x) = Q\left(\frac{x - m}{\sigma}\right)$$

---

7. Worked Probability Transformations

Example: Interval Probability

Given

$$|X+3| < 2$$

Stepwise reasoning:

$$-2 < X+3 < 2$$

Subtracting 3:

$$-5 < X < -1$$

Thus,

$$P(-5 < X < -1) = F_X(-1) - F_X(-5)$$

Using symmetry:

$$= 2\Phi(1) - 1 = 1 - 2Q(1)$$

Example: Two-Sided Probability

Given

$$|X-2| > 1$$

Equivalent to

$$X < 1 \quad \text{or} \quad X > 3$$

Thus,

$$P(X < 1) + P(X > 3)$$

evaluated using Φ and Q.

---

8. Application Examples

Lecture examples:

- Thermal noise in circuits,
- Measurement errors in sensors,
- Packet delay variations in networks.

In each case:

Measured value = True value + Gaussian noise.

---

9. Problem Solving Using a Given CDF

Given

$$F_X(x) = \begin{cases} 0, & x < 0 \\ x^2, & 0 < x < 1 \\ 1, & x > 1 \end{cases}$$

Step 1: Obtain PDF

Differentiate:

$$f_X(x) = 2x, \quad 0 < x < 1$$

Step 2: Compute Mean and Variance

Mean and variance are computed using expectation definitions.

Step 3: Skewness and Kurtosis

Using moment formulas, lecture concludes:

- Distribution is left-skewed.
- Distribution is platykurtic.

---

10. Gaussian Modeling Motivation

Lecture motivation:

Real measurements are noisy; repeated observations create uncertainty clouds.

Model:

$$X = \mu + \sigma Z$$

where Z is standard normal.

Probability models uncertainty rather than exact outcomes.

---

11. Density Estimation

In practice parameters are unknown.

Procedure:

1. Observe noisy samples.
2. Estimate mean and variance.
3. Fit Gaussian distribution.

This replaces raw histogram data with a smooth probabilistic model.

---

12. Networking Application

Packet delay modeling aims to:

- Estimate mean delay,
- Measure jitter.

Observation: large variation harms real-time systems more than constant delay.

---

13. Broader Computing Applications

Gaussian modeling appears in:

- Image denoising,
- Distributed systems,
- Sensor fusion,
- Autonomous systems.

Conclusion: uncertainty modeling is central across computing systems.

Final Revision Checklist

Students should be able to:

1. State Gaussian PDF and parameters.
2. Perform standardization.
3. Use Φ and Q functions correctly.
4. Evaluate interval and tail probabilities.
5. Compute moments from PDF/CDF.
6. Interpret skewness and kurtosis.
7. Explain Gaussian modeling motivation.

End of Lecture 8 Scribe.

Generate a lecture scribe intended to serve as exam reference material. While writing the scribe, reason step by step through definitions, assumptions, proofs, derivations, and example solutions exactly as presented in the provided context. Include explicit reasoning in the scribe so that a student can follow the logical flow during revision. Use only the provided lecture slides and relevant textbook excerpts as context. Maintain a formal academic tone suitable for exam preparation. Follow the structure and scope of the lecture faithfully. Do not introduce any

material, examples, explanations, or assumptions that are not present in the provided context. Prioritize correctness and faithfulness to the lecture material

CSE400 – Fundamentals of Probability in Computing

Lecture 8 Scribe: Gaussian Random Variable — Definition, Standard Forms, Probability Evaluation, and Applications

Instructor: Dr. Dhaval Patel

Course: CSE400 – Fundamentals of Probability in Computing

Lecture Topic: Gaussian Random Variable — Definition and Properties

Purpose

This lecture scribe is prepared strictly from the provided lecture slides and supporting textbook excerpts and is intended to serve as exam-reference material. Definitions, derivations, and example solutions are presented step by step exactly in the sequence developed in the lecture so that students can reproduce the reasoning during revision. No additional assumptions, examples, or external material are introduced.

---

1. Lecture Scope and Logical Development

The lecture develops tools needed to understand and use Gaussian random variables in probability modeling. The structure followed in the lecture is:

1. Review of standardized higher-order moments describing distribution shape.
2. Expected value of functions of random variables.

3. Definition and properties of Gaussian random variables.
4. Standard forms: error function, Φ-function, and Q-function.
5. Relation between Φ and Q functions.
6. Evaluation of Gaussian CDF and tail probabilities.
7. Worked probability examples.
8. Application examples.
9. Problem solving from a given CDF.
10. Motivation for Gaussian modeling.
11. Density estimation from observed data.
12. Networking and broader computing applications.

The lecture emphasizes modeling uncertainty in engineering and computing systems.

---

2. Measures Describing Distribution Shape

Before introducing Gaussian distributions, the lecture revisits measures describing distribution shape.

2.1 Skewness

Skewness measures asymmetry of a distribution around the mean.

It is defined as

$$c_3 = \frac{E[(X-\mu)^3]}{\sigma^3}$$

Reasoning

- The cubic term preserves sign, allowing distinction between left and right deviations.
- Dividing by σ^3 removes units, making comparison possible across distributions.

Interpretation

- Positive skewness → distribution extends more to the right.
- Negative skewness → distribution extends more to the left.
- Zero skewness → symmetric distribution.

Thus, skewness measures directional asymmetry.

2.2 Kurtosis

Kurtosis measures peakedness or flatness of a distribution relative to the Gaussian case.

It is defined as

$$c_4 = \frac{E[(X-\mu)^4]}{\sigma^4}$$

Reasoning

- Fourth power emphasizes large deviations.
- Normalization removes scale dependence.

Interpretation

- Larger values indicate heavier tails and sharper peaks.
- Smaller values indicate flatter distributions.

Thus kurtosis describes concentration around the mean.

3. Expected Value of a Function of a Random Variable

Expectation of a function depends on the type of random variable.

3.1 Continuous Random Variable

For function $g(X)$,

$$E[g(X)] = \int g(x) f_X(x) dx$$

Each value contributes weighted by probability density.

3.2 Discrete Random Variable

$$E[g(X)] = \sum g(x_k) P_X(x_k)$$

Each outcome contributes weighted by probability mass.

3.3 Linearity of Expectation

For constants a and b,

$$E[aX + b] = aE[X] + b$$

Also,

$$E\left[\sum g_k(X)\right] = \sum E[g_k(X)]$$

This property allows expectation to be computed term by term.

---

4. Gaussian Random Variable

4.1 Definition

A random variable is Gaussian if its PDF is

$$f_X(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-m)^2}{2\sigma^2}\right)$$

Notation:

$$X \sim N(m, \sigma^2)$$

where

- m is the mean,
- σ^2 is the variance.

4.2 Observed Properties from Lecture Figures

From lecture graphs:

- Distribution is symmetric about the mean.
- Maximum density occurs at the mean.

- Larger variance produces wider spread.
- The CDF shows smooth S-shaped growth from 0 to 1.

Thus:

- Mean determines location.
- Variance controls spread.

---

5. Standard Forms Used in Gaussian Calculations

To evaluate probabilities easily, Gaussian variables are standardized.

5.1 Error Function

Defined as

$$\operatorname{erf}(x) = \frac{2}{\sqrt{\pi}} \int_0^x e^{-t^2} dt$$

Complementary form:

$$\operatorname{erfc}(x) = 1 - \operatorname{erf}(x)$$

These integrals arise when evaluating Gaussian probabilities.

5.2 Φ-Function (Standard Normal CDF)

Defined as

$$\Phi(x) = \int_{-\infty}^{x} \frac{1}{\sqrt{2\pi}} e^{-t^2/2} dt$$

This gives probability to the left of x.

5.3 Q-Function (Gaussian Tail Function)

Defined as

$$Q(x) = \int_{x}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-t^2/2} dt$$

This gives probability in the right tail.

5.4 Relation Between Φ and Q

Lecture identity:

$$Q(x) = 1 - \Phi(x)$$

Hence,

- Φ is used for CDF evaluation.
- Q is natural for right-tail probabilities.

---

6. Evaluating Gaussian CDF

For $X \sim N(m, \sigma^2)$, define standardized variable

$$Z = \frac{X - m}{\sigma}$$

Then,

$$F_X(x) = \Phi\left(\frac{x - m}{\sigma}\right)$$

Standardization converts the Gaussian variable into standard normal form.

Tail Probability

Right-tail probability becomes

$$P(X > x) = Q\left(\frac{x - m}{\sigma}\right)$$

Thus Q-function directly evaluates right-tail events.

---

7. Worked Probability Transformations

Example: Interval Probability

Given

$$|X+3| < 2$$

Step-by-step reasoning:

$$-2 < X+3 < 2$$

Subtracting 3:

$$-5 < X < -1$$

Hence,

$$P(-5 < X < -1) = F_X(-1) - F_X(-5)$$

Using symmetry relations:

$$= 2\Phi(1) - 1 = 1 - 2Q(1)$$

Example: Two-Sided Condition

Given

$$|X-2| > 1$$

Equivalent to

$$X < 1 \quad \text{or} \quad X > 3$$

Thus,

$$P(X < 1) + P(X > 3)$$

evaluated using Φ and Q.

---

8. Application Examples

Lecture examples include:

- Thermal noise in electronic circuits.
- Measurement error in sensors.
- Packet delay variation in networks.

In each case:

Measured value = True value + Gaussian noise.

---

9. Problem Solving Using a Given CDF

Given

$$F_X(x) = \begin{cases} 0, & x < 0 \\ x^2, & 0 < x < 1 \\ 1, & x > 1 \end{cases}$$

Step 1: Obtain PDF

Differentiate:

$$f_X(x) = 2x, \quad 0 < x < 1$$

Step 2: Compute Mean and Variance

Mean and variance are obtained using expectation formulas.

Step 3: Skewness and Kurtosis

Using moment definitions, lecture concludes:

- Distribution is left-skewed.
- Distribution is platykurtic (lighter tails than Gaussian).

---

10. Gaussian Modeling Motivation

Lecture motivation:

Real-world measurements exhibit randomness; repeated observations create uncertainty clouds.

Model:

$$X = \mu + \sigma Z$$

where Z is standard normal.

Probability describes uncertainty rather than exact outcomes.

---

11. Density Estimation

In practical situations parameters are unknown.

Procedure:

1. Collect noisy samples.
2. Estimate mean and variance.
3. Fit Gaussian distribution.

This replaces raw histogram data with a smooth probabilistic model.

---

12. Networking Application

Packet delay modeling aims to:

- Estimate typical delay.
- Measure jitter (variation).

Observation: large variation harms real-time systems more than constant delay.

---

13. Broader Computing Applications

Gaussian modeling appears in:

- Image denoising,
- Distributed systems,
- Sensor fusion,
- Autonomous systems.

Conclusion: uncertainty modeling is central across computing systems.

Final Revision Checklist

Students should be able to:

1. State Gaussian PDF and parameters.
2. Perform standardization.
3. Use Φ and Q correctly.
4. Evaluate interval and tail probabilities.
5. Compute moments from PDFs and CDFs.
6. Interpret skewness and kurtosis.
7. Explain Gaussian modeling motivation.

End of Lecture 8 Scribe.

Generate a lecture scribe intended to serve as exam reference material. While writing the scribe, reason step by step through definitions, assumptions, proofs, derivations, and example solutions exactly as presented in the provided context. Include explicit reasoning in the scribe so that a student can follow the logical flow during revision. Use only the provided lecture slides and relevant textbook excerpts as context. Maintain a formal academic tone suitable for exam preparation. Follow the structure and scope of the lecture faithfully. Do not introduce any material, examples, explanations, or assumptions that are

not present in the provided context. Prioritize correctness and faithfulness to the lecture material

CSE400 – Fundamentals of Probability in Computing

Lecture 8 Scribe: Gaussian Random Variable — Definition, Standard Forms, Probability Evaluation, and Applications

Instructor: Dr. Dhaval Patel

Course: CSE400 – Fundamentals of Probability in Computing

Lecture Topic: Gaussian Random Variable — Definition and Properties

Purpose

This lecture scribe is prepared strictly from the provided lecture slides and supporting textbook excerpts and is intended to serve as exam-reference material. All definitions, derivations, and example solutions are presented step by step exactly following the lecture progression so that students can reproduce the reasoning during revision. No external material, examples, or assumptions are introduced beyond what appears in the lecture context.

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1. Lecture Scope and Logical Development

The lecture introduces the Gaussian random variable and develops methods for evaluating probabilities involving it. The structure followed in the lecture is:

1. Review of standardized higher-order moments describing distribution shape.
2. Expected value of functions of random variables.

3. Definition and properties of Gaussian random variables.
4. Standard forms: error function, Φ-function, and Q-function.
5. Relation between Φ and Q functions.
6. Evaluation of Gaussian CDF and tail probabilities.
7. Worked probability examples.
8. Application examples.
9. Problem solving using a given CDF.
10. Motivation for Gaussian modeling.
11. Density estimation from observed data.
12. Applications in networking and broader computing systems.

The lecture emphasizes that probability models uncertainty observed in engineering systems.

---

2. Measures Describing Distribution Shape

The lecture begins with measures describing distribution shape beyond mean and variance.

2.1 Skewness

Skewness measures asymmetry of a probability distribution around the mean.

It is defined as

$$c_3 = \frac{E[(X-\mu)^3]}{\sigma^3}$$

Reasoning

- The cubic term preserves sign, allowing distinction between deviations on the left and right of the mean.
- Dividing by σ^3 removes unit dependence.

Interpretation

- Positive skewness → distribution extends further to the right.
- Negative skewness → distribution extends further to the left.
- Zero skewness → symmetric distribution.

Thus skewness quantifies directional asymmetry.

2.2 Kurtosis

Kurtosis measures peakedness or flatness relative to a Gaussian distribution.

Definition:

$$c_4 = \frac{E[(X-\mu)^4]}{\sigma^4}$$

Reasoning

- Fourth power emphasizes large deviations.
- Normalization removes dependence on scale.

Interpretation

- Larger value indicates heavier tails and sharper peak.

- Smaller value indicates flatter distribution.

Thus kurtosis measures concentration of probability around the mean.

---

3. Expected Value of a Function of a Random Variable

Expectation of a function depends on the type of random variable.

3.1 Continuous Random Variable

For function $g(X)$,

$$E[g(X)] = \int g(x) f_X(x)\, dx$$

Each value contributes weighted by probability density.

3.2 Discrete Random Variable

$$E[g(X)] = \sum g(x_k) P_X(x_k)$$

Each outcome contributes weighted by probability mass.

3.3 Linearity of Expectation

For constants a and b,

$$E[aX + b] = aE[X] + b$$

Also,

$$E\left[\sum g_k(X)\right] = \sum E[g_k(X)]$$

This property allows expectations of sums to be computed term by term.

---

4. Gaussian Random Variable

4.1 Definition

A random variable is Gaussian if its PDF is

$$f_X(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-m)^2}{2\sigma^2}\right)$$

Notation:

$$X \sim N(m, \sigma^2)$$

where

- m is the mean,
- σ^2 is the variance.

4.2 Observed Properties

From lecture figures:

- Distribution is symmetric about the mean.
- Maximum density occurs at the mean.
- Larger variance produces wider spread.
- CDF grows smoothly from 0 to 1.

Thus mean controls location and variance controls spread.

---

5. Standard Forms for Gaussian Calculations

To evaluate probabilities easily, Gaussian variables are standardized.

5.1 Error Function

Defined as

$$\mathrm{erf}(x) = \frac{2}{\sqrt{\pi}} \int_0^x e^{-t^2} dt$$

Complementary function:

$$\mathrm{erfc}(x) = 1 - \mathrm{erf}(x)$$

These integrals arise in Gaussian probability calculations.

5.2 Φ-Function (Standard Normal CDF)

Defined as

$$\Phi(x) = \int_{-\infty}^{x} \frac{1}{\sqrt{2\pi}} e^{-t^2/2} dt$$

This gives probability to the left of x.

5.3 Q-Function (Gaussian Tail Function)

Defined as

$$Q(x) = \int_{x}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-t^2/2} dt$$

This gives probability in the right tail.

5.4 Relation Between Φ and Q

Lecture identity:

$$Q(x) = 1 - \Phi(x)$$

Hence:

- Φ is convenient for evaluating CDF values.
- Q is convenient for right-tail probabilities.

---

6. Evaluating Gaussian CDF

For $X \sim N(m, \sigma^2)$, define standardized variable

$$Z = \frac{X - m}{\sigma}$$

Then,

$$F_X(x) = \Phi\left(\frac{x - m}{\sigma}\right)$$

Standardization converts the variable into standard normal form.

Tail Probability

Right-tail probability becomes

$$P(X > x) = Q\left(\frac{x - m}{\sigma}\right)$$

Thus Q-function directly evaluates right-tail events.

---

7. Worked Probability Transformations

Example: Interval Probability

Given

$$|X + 3| < 2$$

Step-by-step reasoning:

$$-2 < X + 3 < 2$$

Subtracting 3:

$$-5 < X < -1$$

Hence,

$$P(-5 < X < -1) = F_X(-1) - F_X(-5)$$

Using symmetry relations,

$$= 2\Phi(1) - 1 = 1 - 2Q(1)$$

Example: Two-Sided Condition

Given

$$|X - 2| > 1$$

Equivalent to

$$X < 1 \quad \text{or} \quad X > 3$$

Thus,

$$P(X < 1) + P(X > 3)$$

evaluated using Φ and Q functions.

8. Application Examples

Lecture examples include:

- Thermal noise in electronic circuits,
- Measurement errors in sensors,
- Packet delay variations in communication networks.

In each case:

Measured value = True value + Gaussian noise.

9. Problem Solving Using a Given CDF

Given

$$F_X(x) = \begin{cases} 0, & x < 0 \\ x^2, & 0 < x < 1 \\ 1, & x > 1 \end{cases}$$

Step 1: Obtain PDF

Differentiate:

$$f_X(x) = 2x, \quad 0 < x < 1$$

Step 2: Mean and Variance

Mean and variance are computed using expectation formulas.

Step 3: Skewness and Kurtosis

Using moment definitions, lecture concludes:

- Distribution is left-skewed.
- Distribution is platykurtic (lighter tails than Gaussian).

---

10. Gaussian Modeling Motivation

Lecture motivation:

Real measurements are noisy and repeated observations create uncertainty clouds.

Model:

$$X = \mu + \sigma Z$$

where Z is standard normal.

Probability models uncertainty rather than exact values.

---

11. Density Estimation

In practice parameters are unknown.

Procedure:

1. Collect noisy samples.

2. Estimate mean and variance.
3. Fit Gaussian distribution.

This replaces raw histogram data with a smooth probabilistic model.

---

12. Networking Application

Packet delay modeling aims to:

- Estimate mean delay,
- Measure delay variation (jitter).

Observation: large variation harms real-time systems more than constant delay.

---

13. Broader Computing Applications

Gaussian modeling appears in:

- Image denoising,
- Distributed systems,
- Sensor fusion,
- Autonomous systems.

Conclusion: uncertainty modeling is central in modern computing systems.

Final Revision Checklist

Students should be able to:

1. State Gaussian PDF and parameters.
2. Perform standardization.
3. Use Φ and Q correctly.
4. Evaluate interval and tail probabilities.
5. Compute moments from PDF/CDF.
6. Interpret skewness and kurtosis.
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End of Lecture 8 Scribe.

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CSE400 – Fundamentals of Probability in Computing

Lecture 8 Scribe: Gaussian Random Variable — Definition, Standard Forms, Probability Evaluation, and Applications

Instructor: Dr. Dhaval Patel

Course: CSE400 – Fundamentals of Probability in Computing

Lecture Topic: Gaussian Random Variable — Definition and Properties

Purpose

This lecture scribe is prepared strictly from the provided lecture slides and supporting textbook excerpts and is intended to serve as exam-reference material. All reasoning, derivations, and example solutions are presented step by step in the same logical sequence as the lecture so that students can reproduce each step during revision. No material outside the provided lecture context is introduced.

---

1. Lecture Scope and Structure

The lecture develops the Gaussian random variable and methods used to compute probabilities involving it. The lecture flow proceeds as follows:

1. Review of higher-order standardized moments describing distribution shape.
2. Expected value of functions of random variables.
3. Definition and properties of Gaussian random variables.

4. Standard forms: error function, Φ-function, and Q-function.
5. Relation between Φ and Q functions.
6. Evaluation of Gaussian CDF and tail probabilities.
7. Worked probability examples.
8. Application examples.
9. Problem solving using a given CDF.
10. Motivation for Gaussian modeling.
11. Density estimation using observed data.
12. Applications in network systems and computing contexts.

The lecture emphasizes modeling uncertainty observed in engineering and computing systems.

---

2. Measures Describing Distribution Shape

The lecture begins by reviewing measures describing distribution characteristics beyond mean and variance.

2.1 Skewness

Skewness measures asymmetry of a distribution about the mean.

Definition:

$$c_3 = \frac{E[(X-\mu)^3]}{\sigma^3}$$

Reasoning

- The cubic term preserves sign, distinguishing left and right deviations.
- Division by σ^3 removes dependence on measurement units.

Interpretation

- Positive skewness → distribution extends further to the right.
- Negative skewness → distribution extends further to the left.
- Zero skewness → symmetric distribution.

Thus skewness quantifies directional asymmetry.

2.2 Kurtosis

Kurtosis measures peakedness or flatness relative to the Gaussian distribution.

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- Fourth power emphasizes large deviations.
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Interpretation

- Larger value indicates heavier tails and sharper peak.
- Smaller value indicates flatter distribution.

Thus kurtosis measures concentration of probability around the mean.

3. Expected Value of a Function of a Random Variable

Expectation of a function depends on whether the random variable is continuous or discrete.

3.1 Continuous Random Variable

For function $g(X)$:

$$E[g(X)] = \int g(x) f_X(x)\, dx$$

Each value contributes weighted by probability density.

3.2 Discrete Random Variable

$$E[g(X)] = \sum g(x_k) P_X(x_k)$$

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For constants a, b:

$$E[aX + b] = aE[X] + b$$

Also,

$$E\left[\sum g_k(X)\right] = \sum E[g_k(X)]$$

Thus expectation distributes across sums and constants, simplifying calculations.

---

4. Gaussian Random Variable

4.1 Definition

A random variable is Gaussian if its PDF is

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where:

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4.2 Observed Properties from Lecture Figures

From lecture graphs:

- Distribution is symmetric around the mean.
- Maximum density occurs at the mean.

- Larger variance produces wider spread.
- CDF increases smoothly from 0 to 1.

Hence:

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Defined as

$$Q(x) = \int_{x}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-t^2/2} dt$$

This gives right-tail probability.

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Lecture identity:

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Thus:

- Φ is used for CDF evaluation.
- Q is natural for right-tail probabilities.

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End of Lecture 8 Scribe.